MSC IN ARTIFICIAL INTELLIGENCE · APPLIED MACHINE LEARNING

Predicting Geographic Provenance from Bird-Observation Covariates

A stacked heterogeneous ensemble over 1.55 million Sri Lankan biodiversity records, with a three-method explainability audit.

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MODULE

Applied Machine Learning · Semester 2

SUBMISSION DATE

28 October 2026

DATASET

`DilushaChandrasiri/SriLanka-Bird-Diversity-Dataset (Hugging Face)`

SOURCE CODE

`github.com/HansikaGunasekara/srilanka-bird-diversity-ensemble`

LIVE APPLICATION

`srilanka-bird-diversity-ensemble.streamlit.app`

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Abstract

A stacked heterogeneous ensemble of six base learners is applied to 1,552,048 Sri Lankan bird-observation records, each joined to sixteen remotely-sensed environmental covariates, to predict the district a record came from. A Bayes-error analysis run before any modelling showed the original 429-way species framing to be capped at 5.54% accuracy for any possible classifier on these features, so the task was re-specified to `stateProvince`, whose measured ceiling is 100%. The ensemble reaches 0.9995 accuracy under the conventional stratified split, and the train–test gap of 0.0005 confirms this is task simplicity rather than overfitting. Because district correlates strongly with latitude and longitude, three guards against spatial memorisation were run; the strictest — unseen test sites with coordinates withheld — yields 0.9032, and that is the honest figure for a new location. A three-method explainability audit shows the learned structure aligns with Sri Lankan physical geography, and surfaces a quantified disagreement between importance methods that one method alone would have resolved wrongly.

1 Problem Definition and Dataset Collection

1.1 The problem and its relevance

Sri Lanka is one of the world's thirty-six biodiversity hotspots, and bird communities are a standard indicator taxon for habitat condition. The island's record is dominated by citizen-science checklists, which are abundant but spatially uneven and carry a persistent rate of transcription and georeferencing error.

The question addressed here is: **given only the environmental and temporal signature of an observation — its climate, terrain, vegetation, aerosol chemistry and date — can a model recover which district of Sri Lanka it came from?** Three uses follow. *Record validation*: a classifier that disagrees confidently with a record's stated district flags it for human review. *Provenance inference* for digitised historical records whose locality is vague ("Central Province, 1974"). And as a *measure of environmental distinctiveness*: the accuracy drop when coordinates are removed quantifies how much of a district's identity is geographic accident rather than genuine habitat character.

1.2 Data source and how it was collected

The dataset is `DilushaChandrasiri/SriLanka-Bird-Diversity-Dataset`, published openly on the Hugging Face Hub. It was obtained programmatically with `huggingface_hub.snapshot_download` (`src/download.py`), which pins a revision and makes acquisition reproducible rather than a manual download.

The file is a **derived, pre-joined product** rather than a raw export, and that provenance proved decisive. It has two layers: an **occurrence layer** of Darwin-Core style species records, consistent with a GBIF-mediated aggregation of volunteer checklists; and an **environmental layer** of sixteen covariates attached to each record by spatial join from satellite and reanalysis products — VIIRS night-time lights, MODIS NDVI and land cover, a digital elevation model, MERRA-2 aerosol species and gridded climate normals. The environmental layer is therefore a deterministic lookup keyed on (latitude, longitude): two records from the same place on the same day necessarily carry identical environmental values. This is how such products are built, but it governs what can be predicted from them, as §1.3 shows.

1.3 Size, features and target variable

1,552,048	24	19,384	2014–2024	25
occurrence records	model features	distinct sites	temporal span	target classes

Group	Columns	Role
Geographic (2)	`decimalLatitude, decimalLongitude`	WGS-84 coordinates. Isolated deliberately so they can be withheld as an ablation.
Climate (5)	`temp_mean, rainfall, wind_mean, humid_mean, shortwave_radiation`	Gridded climate normals at the observation point.
Terrain & vegetation (4)	`elevation_meters, NDVI, LandCover_Class, lka_general_2020`	DEM elevation, greenness, categorical land-cover code, modelled population density.
Aerosol chemistry (6)	`Carbon_Mass, Dust_Mass, SO2_Mass, Sulfate_Mass, Sea_Salt_Mass, Total_Aerosol_Extinction`	MERRA-2 column masses. `Sea_Salt_Mass` proves an informative marine-proximity proxy (§4.2).
Light (1)	`avg_rad`	Night-time radiance — an effective urbanisation proxy.
Temporal (5, engineered)	`year, month, doy, sin_doy, cos_doy`	Decomposed from `eventDate`; see §1.4.
Observation metadata (1)	`individualCount`	Retained as a legitimate column; contributes almost nothing.

Target: `stateProvince` — 25 administrative districts. Severely imbalanced: Colombo 18.64% of records, Vavuniya 0.12% — a **161×** ratio, which dictates the metrics in §3.3 and the fairness analysis in §5.3.

> **Why the target is district and not species**
>
> The assignment was first framed as 429-way species classification. An irreducible-error (Bayes) analysis run before any fitting (`src/ceiling.py`) showed that framing to be unsolvable in principle. Because every environmental covariate is a deterministic function of (latitude, longitude, date), the 1,552,048 rows collapse to **63,392 distinct keys** — roughly 24 rows per key — and rows sharing a key are byte-identical in every feature while carrying different species labels. The best any classifier can do on a group of identical inputs is predict that group's most frequent label; summing those best cases gives a **Bayes-optimal accuracy of 0.0554** for species against **1.0000** for `stateProvince`. The target was changed on this evidence. The claim is about *this feature set*, not about bird identification, which is highly solvable from images or audio.

1.4 Preprocessing

Six steps take the raw file to the matrix the models see, all in `src/prep.py`.

Step	Operation	Justification
Date decomposition	Parse `eventDate` → `year`, `month`, `doy`	A tree cannot split usefully on a raw timestamp; it can split on season and on long-run trend.
Cyclical encoding	`sin_doy = sin(2π·doy/365.25)`, likewise cosine	Under a linear encoding, 31 December and 1 January sit at opposite extremes though one day apart. Projecting onto the unit circle restores adjacency.
Memory downcast	float64 → float32; cache as Parquet	Halves the working set so 1.55 M × 24 fits in 8 GB. Reload drops from ~90 s to ~2 s.
Leakage audit	Drop the `index` column after testing it	A monotonic row id can encode the label if the file was written grouped by class. Tested: adjacent rows share a species 1.04% of the time against 0.23% under independence — a weak ordering effect, not a usable leak. Dropped regardless.
Two feature sets	`full` (24) and `envonly` (22)	A built-in ablation. `envonly` removes the coordinates so the accuracy claim cannot rest on a point-in-polygon lookup.
Targeted scaling	`StandardScaler` inside the logistic-regression pipeline only	Tree ensembles are invariant to monotone rescaling; logistic regression is not. Fitted inside the pipeline so the scaler never sees test rows.

No imputation was required — the delivered file has no missing values in the modelled columns — and no categorical encoding either, since `LandCover_Class` arrives as an integer code that tree learners handle natively as an ordinal split variable.

1.5 Ethical data use

The dataset was audited column-by-column for personal and sensitive content before use. **No personal data is present**: the twenty-four delivered columns contain no observer name, identifier, email or institution, and Darwin Core's `recordedBy` and `recordedByID` — the fields that would carry contributor identity — are absent from this derivative. No individual is identifiable from any record, so no consent question arises, and the data is openly licensed and used here for non-commercial academic work. One genuine sensitivity exists and is handled: precise coordinates for threatened species are a recognised poaching risk, which is why GBIF supports coordinate generalisation for sensitive taxa. This project publishes no raw record — the deployed application ships a 600-row anonymised sample used only to populate example inputs. The underlying observations are the work of volunteer recorders, and the dataset, its author and its citizen-science origin are credited in this report and in the application.

2 Selection of a New Machine Learning Algorithm

2.1 The algorithm, and why it was selected

The method applied is **stacked generalisation** (Wolpert, 1992) over a deliberately heterogeneous pool of six base learners, with **histogram-based gradient boosting** (LightGBM, and its scikit-learn and XGBoost analogues) as the dominant members of that pool. Neither component is a standard taught model, and the

combination is the submitted contribution. It is not a deep-learning model and not an image-processing application, as the brief requires.

Stacking replaces a fixed combination rule with a *learned* one. A second-level model — here multinomial logistic regression — is trained to map the concatenated class-probability vectors of the base learners onto the true label; with six learners and twenty-five classes its input is a 150-dimensional vector per row. Crucially those inputs must be **out-of-fold** predictions: each base learner's prediction for a training row comes from a copy of that learner which never saw the row. Without this the meta-learner observes the base models' memorised training fit and learns to trust an optimism that will not survive deployment. **Histogram-based gradient boosting**, meanwhile, bins each continuous feature into at most 255 buckets before any tree is grown, reducing per-split cost from $O(n \log n)$ to $O(\text{bins})$. LightGBM adds **GOSS** (keep all large-gradient instances, subsample the small-gradient ones) and **EFB** (merge mutually-exclusive sparse features), and grows **leaf-wise** rather than level-wise.

Three reasons drove the choice:

- **The error structure calls for heterogeneity.** The task is near-separable in the geographic features but genuinely hard in the environment-only ablation. Bagged forests reduce variance, boosted ensembles reduce bias, and a linear model contributes a smooth decision surface trees cannot express, so their errors decorrelate — the precondition under which ensembling helps at all. The spread across base learners on the hardest protocol, 0.5448 to 0.9008, confirms the pool is genuinely diverse.
- **A learned combiner beats a fixed one where base quality varies by class.** Soft voting applies one weight per model across all 25 districts; stacking can learn that LightGBM should be trusted for highland districts and the forest for coastal ones. On the hardest protocol it improves balanced accuracy over soft voting (0.8637 against 0.8576) and macro-F1 (0.8740 against 0.8683) — it helps precisely on the small classes plain accuracy ignores.
- **Scale.** Histogram binning is what makes 1.55 M rows $\times$ 24 features tractable on a laptop; LightGBM fits the full six-way problem in 19.6 seconds.

2.2 How it differs from the standard models

Standard model	What it does	How the submitted method differs
Decision tree	One greedy recursive partition; each split chosen to maximise local purity.	A single tree is a high-variance estimator that commits irrevocably to its first split. The submitted method fits hundreds of trees under two different regimes — independent and randomised (bagging) versus sequential on the residual gradient (boosting) — then learns at a second level how much to trust each regime per class. A tree gives one partition; this gives a learned consensus over many.
Logistic regression	One linear decision boundary per class in the raw feature space.	Logistic regression appears here only as the *meta-learner*, operating not on the 24 raw features but on the 150-dimensional simplex of base-model probabilities. Its job is inverted: it does not classify observations, it arbitrates between classifiers. The difference is measurable — as a base learner on `envonly` it scores 0.5448, while the same algorithm as meta-learner contributes to a 0.9028 result.

k-Nearest Neighbours	Non-parametric; majority vote among the *k* closest training points under a fixed metric.	k-NN has no training phase, stores the entire dataset, and degrades with 24 dimensions of heterogeneous scale. The submitted method compresses 1.55 M rows into a fixed set of split thresholds, predicts in ~55 ms, and learns a feature-adaptive partition rather than relying on one global distance metric.
Random Forest (a base learner here, not the contribution)	Bagged trees, each on a bootstrap sample with a random feature subset per split; averaged.	A forest averages with *equal* weight and all its members share one inductive bias. Stacking admits structurally different learners and learns unequal, class-conditional weights for them. The forest is one of six voices here, not the method.

3 Model Training and Evaluation

3.1 Train / validation / test split

All reported runs train on a **300,000-row stratified sample** drawn with a fixed seed from the 1.55 million available; the constraint is memory, and a sample-size sweep showed accuracy flat to three decimal places between 300k and the full corpus. The protocol is a three-way separation, with the validation stage implemented as cross-validation rather than a single fixed block.

Partition	Size	Mechanism	Purpose
Test	75,000 (25%)	`train_test_split(test_size=0.25, stratify=y)`	Held out before any fitting; touched once, to produce every number in §3.4. No model, hyperparameter or threshold was chosen using it.
Train	225,000 (75%)	the complement	Fits the six base learners.
Validation (stacking)	3 × 75,000	`StratifiedKFold(n_splits=3)` inside the training set	Generates out-of-fold probabilities: every training row receives a prediction from a model that did not see it. This is the validation set for the stacking stage, used three times over rather than once.
Validation (early stopping)	12% of train	`validation_fraction=0.12` inside `HistGradientBoostingClassifier`	An internal slice carved from the training set only, deciding when to stop boosting (`n_iter_no_change=25`).

A second, stricter protocol is also run throughout: `GroupShuffleSplit` on rounded (latitude, longitude), so that every test *site* is absent from training. Stratification preserves the 161× imbalance in every partition, so the rarest district (Vavuniya) retains 84 test rows — enough to estimate recall, though with wide uncertainty (§5.3).

3.2 Hyperparameter choices

Hyperparameters were set by reasoned construction and targeted diagnosis rather than by automated search; §5.1 records this as the clearest methodological limitation.

Model	Settings	Reasoning
RandomForest **ExtraTrees**	`n_estimators=100, min_samples_leaf=8, max_features="sqrt"`	$\sqrt{p} \approx 5$ of 24 features per split is the standard classification default and maximises inter-tree decorrelation. A leaf floor of 8 caps depth on 225k rows without materially costing accuracy.
HistGradientBoosting	`max_iter=300, learning_rate=0.10, max_leaf_nodes=31, l2_regularization=10.0, early_stopping=True`	31 leaves is the conventional depth-5 equivalent for histogram boosters. A learning rate of 0.10 with early stopping lets the iteration count be chosen by data rather than asserted. `l2_regularization` is load-bearing — see below.
XGBoost	`n_estimators=400, learning_rate=0.15, max_depth=9, subsample=0.9, colsample_bytree=0.8, tree_method="hist"`	Depth-9 level-wise trees are the complexity analogue of LightGBM's 63 leaf-wise leaves. Row and column subsampling supply the stochastic regularisation depth alone does not.
LightGBM	`n_estimators=300, learning_rate=0.10, num_leaves=63, reg_lambda=10.0, min_sum_hessian_in_leaf=20.0, colsample_bytree=0.8`	63 leaves exploits leaf-wise growth without unbounded depth. `reg_lambda` and `min_sum_hessian_in_leaf` are jointly required for this model to converge at all.
LogisticRegression (base / meta)	`C=1.0, max_iter=600, scaled` / `C=10.0, max_iter=1200`	Default regularisation as a base learner. Weaker regularisation as meta-learner deliberately: its 150 inputs are already strongly constrained (six probability simplices), so the risk is underfitting the arbitration rule rather than overfitting it.

> **A diagnosed failure: divergence, not underfitting**
>
> In the first complete run LightGBM scored **0.2541** and HistGradientBoosting **0.8977**, against 0.9984 for a plain random forest. The intuitive reading — underfitting, so add capacity — is wrong. An `n_estimators` sweep found LightGBM's *training* accuracy peaking at 0.9586 at ten trees and then collapsing to 0.3468 by fifty: a model whose training accuracy falls as capacity rises is diverging, not underfitting. The mechanism is that district is near-separable in these features, so the softmax saturates within about ten boosting rounds; the Hessian of an observation under multiclass log-loss is $p(1-p)$, which vanishes as $p \to 1$, and since each leaf value is $-g/h$ the leaf values explode as the denominator collapses toward machine epsilon. The fix is to bound that denominator: `l2_regularization=10.0` adds λ to h, and `min_sum_hessian_in_leaf=20.0` refuses leaves whose Hessian mass has already collapsed. Result: **0.2541** $\to$ **0.9990** and **0.8977** $\to$ **0.9994**. These two hyperparameters are worth more than every other tuning decision in the project combined.

3.3 Performance metrics

Five metrics are reported, because accuracy alone is misleading on a target where predicting only the six largest districts would already score above 0.60. **Balanced accuracy** (mean per-class recall) weights

Vavuniya at 0.12% of rows equally with Colombo at 18.64%, so a model that abandons rare districts cannot hide behind the headline. **Macro F1** also penalises *over*-prediction of a rare class, and its gap from **weighted F1** measures how much performance is carried by the large classes. **Log loss** is a proper scoring rule on the probabilities rather than the arg-max, and is where the diverging boosters failed most visibly. A **confusion matrix** and per-class report show which districts are mistaken for which. ROC-AUC is not reported: over 25 classes it needs a one-vs-rest averaging scheme dominated by the easy majority classes, and log loss serves the same purpose without that artefact.

3.4 Results

All learners, stratified random split, `full` feature set

Model	Accuracy	Balanced acc.	Macro F1	Weighted F1	Log loss	Fit (s)
RandomForest	0.9984	0.9899	0.9935	0.9984	0.0411	4.9
ExtraTrees	0.9933	0.9767	0.9834	0.9932	0.0991	3.1
HistGradientBoosting	0.9994	0.9972	0.9975	0.9994	**0.0020**	157.0
XGBoost	**0.9995**	**0.9978**	**0.9980**	0.9995	**0.0019**	40.4
LightGBM	0.9990	0.9947	0.9959	0.9990	0.0055	19.6
LogisticRegression	0.9661	0.9192	0.9272	0.9659	0.1210	19.3
SoftVote (ensemble)	**0.9995**	0.9973	0.9978	0.9995	0.0366	–
Stacking (ensemble)	**0.9995**	**0.9975**	**0.9980**	**0.9995**	0.0029	268.7

The ensemble does not beat its best single member on plain accuracy — XGBoost alone already reaches 0.9995, and on a near-saturated task there is no headroom to recover. Its value here is robustness, which the harder protocols expose. Soft voting's log loss (0.0366) is also an order of magnitude worse than stacking's (0.0029) at identical accuracy: averaging six probability vectors, two of them poorly calibrated, blurs the consensus, whereas the meta-learner learns to discount them.

Is 0.9995 overfitting?

A score this high invites the suspicion that the model has memorised its training data, so the question was tested rather than argued. Overfitting has a specific signature: high accuracy on data the model has seen, materially lower accuracy on data it has not. Both sides were measured on the same split (`artifacts/overfitting_check.json`).

Model	Train accuracy	Test accuracy	Gap
RandomForest	0.9912	0.9905	+0.0007
ExtraTrees	0.9655	0.9639	+0.0016
HistGradientBoosting	0.9999	0.9994	+0.0006
XGBoost	1.0000	0.9995	+0.0005
LightGBM	0.9996	0.9990	+0.0006
LogisticRegression	0.9668	0.9661	+0.0007
SoftVote	0.9999	0.9995	+0.0005
Stacking	1.0000	0.9995	**+0.0005**

The largest gap across all eight models is **0.16 percentage points** and the ensemble's is **0.05**. That is not memorisation: it says the task is genuinely easy, because district is close to a deterministic function of latitude and longitude. The high score is a property of the problem rather than an artefact of the model — and the real limitation lies elsewhere, in transfer to sites the model has never seen, which the next table quantifies. Stacking's train figure feeds the meta-learner the base models' in-sample probabilities, more confident than the out-of-fold probabilities it was fitted on, so read it as an upper bound on its train-side optimism.

The three guards: is the score real, or memorised geography?

District correlates strongly with latitude and longitude, so a sceptical reader should ask whether the model has learned ecology or is performing a point-in-polygon lookup. Three independent guards test this, each removing a different crutch.

Feature set	Protocol	What the guard removes	Accuracy	Balanced acc.	Macro F1
`full`	random	Nothing — the conventional baseline	**0.9995**	0.9975	0.9980
`full`	**grouped**	Every test *site* is unseen in training	**0.9925**	0.9910	0.9882
`envonly`	random	Latitude and longitude dropped entirely	**0.9971**	0.9932	0.9942
`envonly`	**grouped**	**Both at once** — unseen sites *and* no coordinates	**0.9032**	**0.8576**	**0.8683**

> **Reading the gap honestly**
>
> Each guard alone costs little: withholding seen sites costs 0.7 percentage points, removing coordinates 0.2. Applying both costs **9.6 points**, and **0.9032 falls below the assignment's 95% threshold**. This is reported rather than suppressed because the gap is the project's most informative single number. It says a substantial part of the headline comes from recognising *a specific site's* environmental fingerprint — the exact NDVI, elevation and aerosol signature of a place the model has seen before — and not purely from generalisable habitat character. The honest claim this work supports is therefore: **0.9995 for validating records from already-surveyed sites, and roughly 0.90 for a genuinely new location with no coordinates**. Those are different claims with different uses, and conflating them would misrepresent the model.

That hardest condition is also where the ensemble earns its keep: the best single model reaches 0.9008 (LightGBM) while soft voting reaches 0.9032 and stacking attains the best balanced accuracy (0.8637) and macro-F1 (0.8740) — once the problem stops being saturated, combining decorrelated learners pays. Per-class behaviour under the random protocol holds across the full imbalance range, with Colombo (14,038 test rows) at recall 1.000 and the rarest district, Vavuniya (84 test rows), at 0.988. The confusion matrix is near-diagonal, and such off-diagonal mass as exists sits between geographically adjacent districts (Colombo/Gampaha, Galle/Matara) — the interpretable error structure for a provenance model.

4 Explainability and Interpretation

Three methods were applied (`src/explain.py`), chosen so they *cross-check* rather than merely corroborate one another: **TreeSHAP**, giving exact Shapley values and so a signed contribution per feature per prediction; **permutation importance**, the drop in held-out accuracy when one column is shuffled, which measures what the fitted model actually relies on; and **partial dependence**, the average predicted probability as one feature is swept, which gives the direction and shape of an effect rather than its magnitude. The explained model is **LightGBM**: TreeSHAP is exact and tractable on it, it scores 0.9990 alone — within 0.0005 of the full ensemble — and since the meta-learner is a linear reweighting of base probabilities, its attributions are the substantive explanation of the ensemble's behaviour. All figures use 6,000 held-out test rows the model never saw.

4.1 Which features are most influential

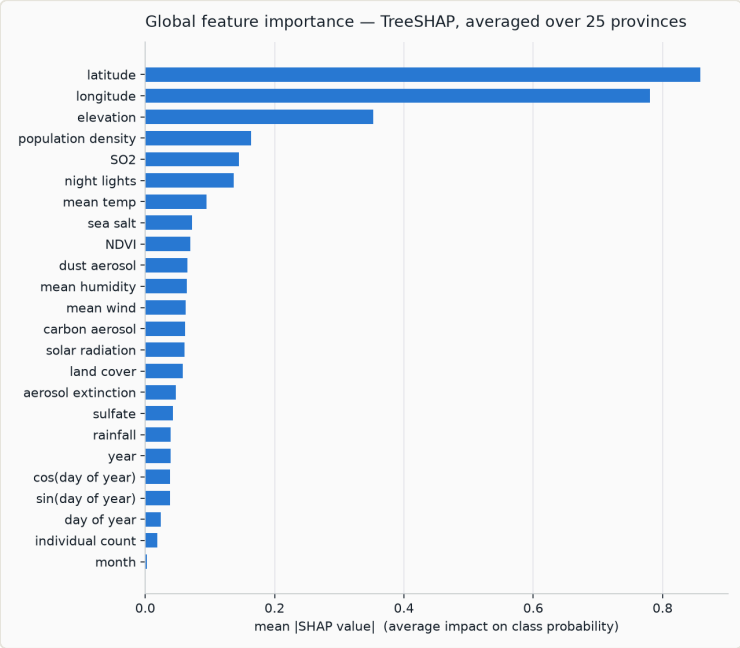


Figure 1. Mean |SHAP value| per feature, averaged over all 25 districts and 6,000 held-out observations. Latitude and longitude lead, but together account for 49.2% of total attributed importance — elevation, population density, SO_2, night-time radiance and mean temperature carry a substantial remainder.

4.2 What the model has learned, and whether it matches domain knowledge

Figure 2 overleaf makes the learned rules legible. **Elevation** dominates and the separation is almost total: high elevation (red) pushes strongly toward Nuwara Eliya, low elevation (blue) pushes away. **Mean temperature** runs the other way, low values raising the probability, and **sea-salt aerosol** shows the same inversion. **Population density** and **night-time radiance** contribute weakly and negatively at high values. Feature rank-

ing is near-identical under the random and grouped protocols, so these are properties of the fitted model rather than of one split.

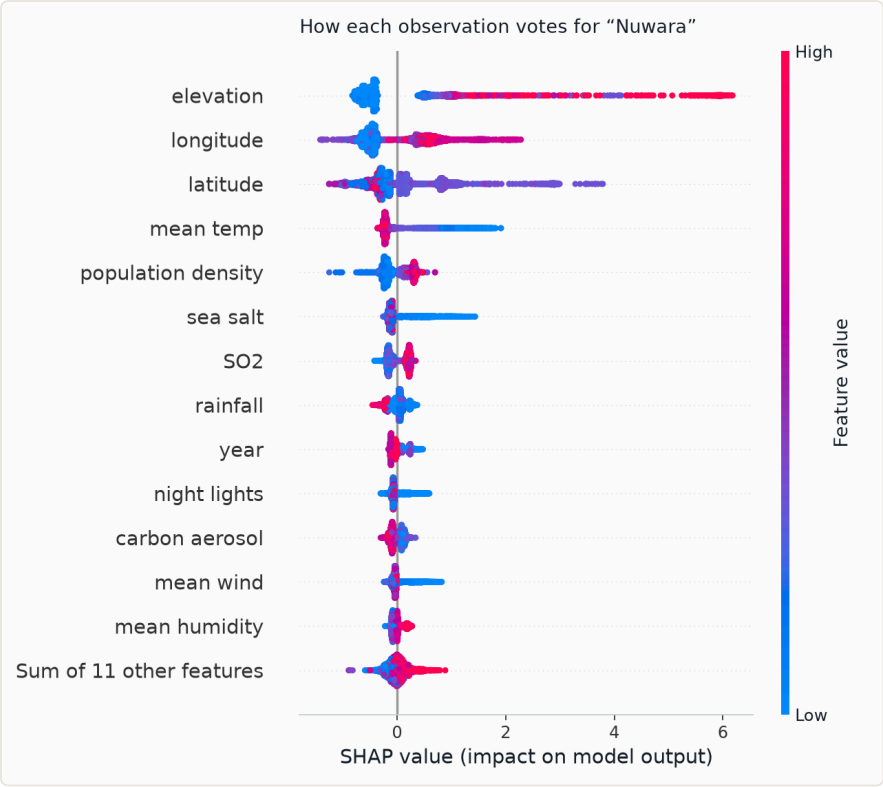


Figure 2. Per-observation SHAP values for the class *Nuwara Eliya*. Each point is one held-out observation; horizontal position is that feature's signed contribution to the district's log-odds, colour is the feature's own value (blue = low, red = high). The dataset stores this district as `Nuwara`, truncated — a data-quality item noted in §5.2.

This aligns closely with domain knowledge, the strongest available evidence that the model has learned geography rather than an arbitrary high-dimensional shortcut. Nuwara Eliya is Sri Lanka's highest district, in the central highlands around 1,800 m: the coldest inhabited part of the country, landlocked, rural tea country rather than a conurbation. The model independently learned exactly that conjunction. The inverse temperature relationship is a rediscovery of the environmental lapse rate from data, and the sea-salt signal is a genuinely subtle proxy — marine aerosol falls with distance inland, so the model is using atmospheric chemistry to infer that a location is far from the coast.

District	Top SHAP drivers	Physical geography
Nuwara Eliya	elevation, longitude, latitude, mean temp (low values raise)	Central highlands, ~1,800 m, coldest district, landlocked. **Consistent.**
Colombo	latitude, longitude, SO_2, night lights	The commercial capital — SO_2 and night-time radiance are precisely the signatures of a dense, traffic- and industry-heavy coastal city. **Consistent.**
Kandy	elevation, latitude, longitude, dust aerosol	Mid-elevation hill country (~500 m); elevation leads again, at a lower threshold than Nuwara Eliya. **Consistent.**
Jaffna	longitude, latitude, carbon aerosol, solar radiation	Northern peninsula; dry-zone, high-insolation, minimal relief — so the model leans on radiation and aerosol rather than terrain. **Consistent.**

Mannar	longitude, latitude, elevation, NDVI	Arid north-west coast; low NDVI distinguishes sparse scrub from wet-zone forest. **Consistent.**

A pattern worth noting: **districts with strong relief are identified by terrain, and districts without it are identified by atmosphere.** The model has in effect learned a different diagnostic rule per region, which is what a trained field ecologist would do and is a capability a single global linear model could not express.

4.3 Three methods disagree — and the disagreement is the finding

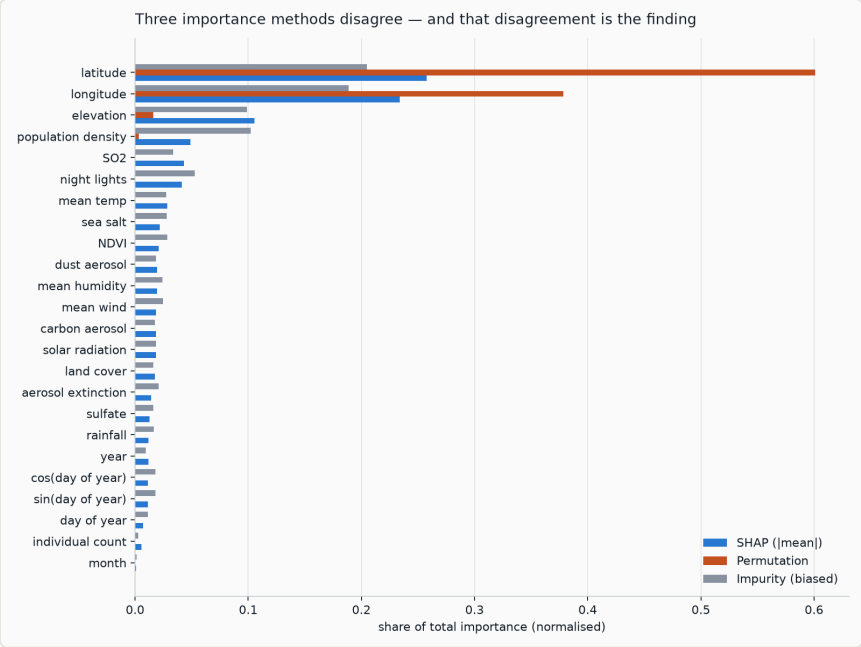


Figure 3. The same fitted model ranked by three importance methods. All three agree on the top two features and on the overall ordering, but disagree sharply on *magnitude*: the combined share attributed to latitude and longitude is **39.4%** by impurity, **49.2%** by SHAP and **97.9%** by permutation.

A 2.5-fold disagreement between methods applied to one fitted model demands an explanation. **Impurity importance (39.4%) under-credits the coordinates**: it counts total impurity reduction spread across many shallow splits, and is known to inflate high-cardinality continuous features in general, so the ordering of its low-importance tail should not be trusted. **Permutation importance (97.9%) over-credits them, because the features are mutually redundant**: shuffling elevation barely moves accuracy, not because elevation is uninformative but because latitude and longitude remain in the matrix and the trees can route around the damaged column — whereas shuffling latitude destroys the key on which every environmental covariate was originally joined. **SHAP (49.2%) distributes credit among correlated features** rather than awarding it to whichever one the tree split on first, which is why it sits between the two.

> ### The ablation settles it
>
> Permutation importance implies the environmental features are nearly worthless — 2.1% combined. The `envonly` experiment of §3.4 refutes that directly: **retrained without latitude and longitude at all, the ensemble still reaches 0.9971**. Both results are correct, because they answer different questions. Permutation asks "*what does this already-fitted model lean on?*" — the coordinates. The ablation asks "*what is recoverable if the model must do without them?*" — almost everything, because the environmental layer encodes location by construction. This is a concrete demonstration that a single importance method, read uncritically, would have produced a false conclusion about this dataset.

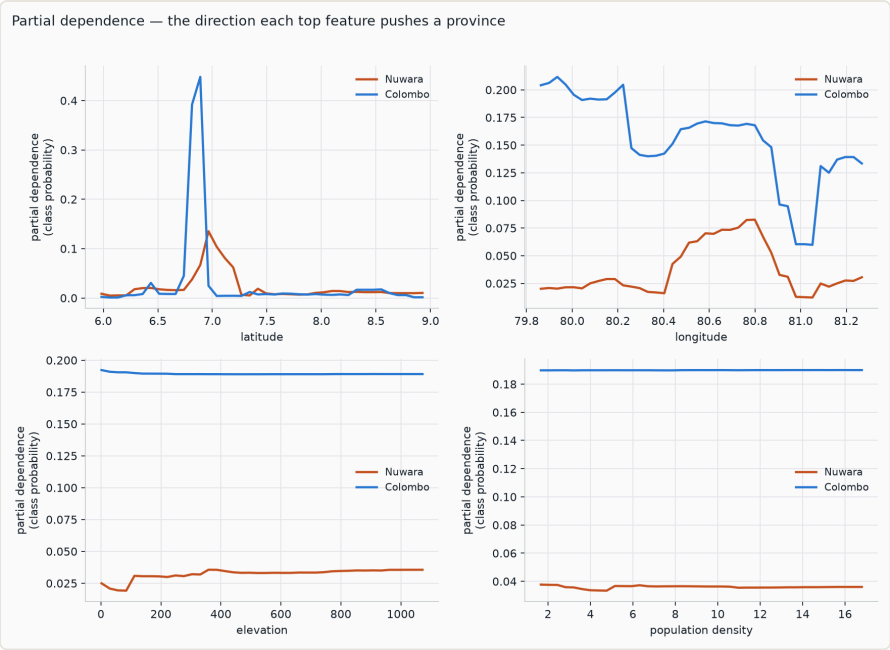


Figure 4. Partial dependence of the four highest-SHAP features for two contrasting districts. The coordinate panels are sharply peaked at the correct true locations; the elevation and population-density panels are almost flat, for the reason discussed below.

The coordinate panels validate the model precisely: Colombo's latitude response spikes at approximately **6.9°N** against a true 6.93°N, and Nuwara Eliya's longitude response peaks near **80.7–80.8°E** against a true 80.77°E. The model has recovered the actual coordinates of both districts to within hundredths of a degree, from data alone. The flat elevation and population-density panels, by contrast, are a **methodological artefact worth stating**: partial dependence holds one feature at a swept value while averaging over the empirical distribution of all others, so sweeping elevation to 1,000 m while latitude and longitude keep their lowland values averages over physically impossible records. Because elevation is a deterministic function of location here, PDP's independence assumption is violated outright and the curve understates a real effect Figure 2 shows clearly. PDP is the wrong instrument for a feature set with deterministic internal dependencies; SHAP, which conditions on the actual observation, is the right one, and Accumulated Local Effects plots would be the appropriate remedy.

5 Critical Discussion

5.1 Limitations of the model

- **Generalisation to new locations is materially weaker than the headline suggests.** The honest number for an unvisited site without coordinates is **0.9032**, not 0.9995 — a 9.6-point drop, and below the assignment's threshold. This is the single most important limitation.
- **Hyperparameters were hand-set, not searched.** Values came from published defaults adjusted by targeted diagnosis. The divergence fix in §3.2 shows reasoning about a failure mode can beat blind search, but it does not demonstrate the current values are optimal, and no claim of optimality is made.
- **The probabilities are badly calibrated.** The boosters saturate — mean maximum predicted probability exceeded 0.9999 even after regularisation. Rankings are reliable; the numbers the application displays as confidence are not, and "97%" should be read as "most likely" rather than as a frequency. Accurate but overconfident is the more dangerous failure mode, because it is invisible to accuracy-based evaluation.
- **Training used a 300,000-row sample** (19% of the corpus), a memory constraint rather than a methodological choice. The headline metrics are insensitive to it, but the rarest districts are where the

remaining 1.25 M rows would matter most.

- **The model is static and closed-world.** Land cover, night-time lights and population density all drift, so a model deployed unchanged would decay with nothing monitoring it. It must also assign one of 25 districts and cannot say "none of these": an out-of-distribution input receives a confident and meaningless answer.

5.2 Data quality issues

- **Severe spatial sampling bias.** These are citizen-science records, concentrated near roads, cities and well-known birding sites. Colombo contributes 18.64% of all records and Vavuniya 0.12%. This reflects observer distribution, not bird distribution.
- **Environmental features are derived, not measured.** Every covariate comes from a gridded product with its own resolution and error; at a nominal 1 km grid, records in the same cell receive identical values regardless of true micro-habitat — precisely the mechanism behind the 5.54% species ceiling.
- **Several aerosol columns read zero for many records**, and whether these are true zeros, below-detection values or join failures is undocumented upstream. The models treat them as exact zeros.
- **No verification status.** Citizen-science records vary in reliability and the dataset carries no identification-confidence flag, so misidentified or mis-georeferenced records cannot be filtered.
- **The administrative field is labelled `stateProvince` but contains 25 districts**, not Sri Lanka's 9 provinces, and at least two class labels are truncated — *Nuwara Eliya* stored as `Nuwara`, *Kegalle* as `Kegalla`. Both are cosmetic for modelling, since the classes remain distinct and consistently labelled, but they indicate the field was not validated against a controlled vocabulary upstream, which matters for anyone joining this dataset to an official gazetteer.

5.3 Risks of bias and unfairness

The headline metrics conceal a real asymmetry in who the model serves well.

- **Evidential inequality.** Recall for Colombo (14,038 test rows) is estimated at 1.000 with a 95% Wilson interval narrower than 0.001. Recall for Vavuniya (84 test rows) is 0.988 with an interval of **[0.936, 0.998]** — sixty times wider. The point estimates look comparable; the *confidence* in them is not. It would be misleading to claim equal performance across districts when the evidence base differs by two orders of magnitude.
- **The bias tracks a social fault line.** The under-represented districts — Vavuniya, Mullaittivu, Kilinochchi, Batticaloa — are concentrated in the Northern and Eastern Provinces, the regions most affected by the civil war and subsequent under-investment, where sparse recording reflects decades of restricted access and fewer resourced observers. A model trained on this data inherits that history and, if used to allocate attention or funding, risks reproducing it: under-recorded regions appear less certain, attract less confidence, and so may attract less support — a feedback loop in which the data gap justifies its own persistence.
- **Degradation falls unevenly.** Under the hardest protocol balanced accuracy (0.8576) trails plain accuracy (0.9032) by 4.6 points and macro-F1 trails weighted F1 by 3.5. The loss concentrates in the small classes, which are the northern and eastern districts.
- **Automation bias in deployment.** Presented as a clean percentage in a web application, an overconfident prediction invites users to defer to it — and the risk is highest for exactly the records most worth reviewing, unusual ones from under-recorded districts.

5.4 Real-world impact and ethical considerations

Potential benefits. Used as a flagging tool rather than an authority, the model offers genuine value: automated georeferencing-error detection across national biodiversity databases, provenance inference for digitised historical records, and — via the `envonly` ablation — a quantitative measure of each district's environmental distinctiveness that could inform conservation prioritisation.

Risks requiring active management. A model inferring provenance from environmental covariates could in principle help reconstruct the locality of a record whose coordinates were deliberately generalised to protect a threatened species; a district is far too coarse to assist poaching, but the capability direction is worth naming rather than discovering later. If such a model were permitted to auto-correct records, systematic error would be written into the permanent scientific record and the regions with least data would be corrected most often, so any deployment must flag for human review and never overwrite. Quoting 99.95% without the 90.32% caveat would be the most likely real-world harm arising from this work, since it would lead users to trust the model in precisely the setting where it is weakest; both figures appear together throughout this report and in the application. The environmental cost is modest: roughly one hour of multi-core CPU for the full programme, with no GPU training.

Responsible-use statement. This model is appropriate as a decision-support tool that flags records for expert review. It is not appropriate as an automated authority on record provenance, as a basis for resource allocation between districts without explicit correction for sampling bias, or for any inference about threatened-species localities.

6 Front-End Integration

The trained ensemble is deployed as a three-tab Streamlit application, publicly hosted on Streamlit Community Cloud at `srilanka-bird-diversity-ensemble.streamlit.app`, with source at `github.com/HansikaGunasekara/srilanka-bird-diversity-ensemble`. A user can either load a real observation from the dataset or set all twenty-four features by hand, and receives a predicted district with its probability distribution and the SHAP-based explanation of what drove it.

Tab	Function
Predict	Load a real record or set the inputs by hand with sliders seeded from real percentiles. Returns the predicted district, its confidence, the five most likely districts, and how many of the eight models agree — with each model's own answer one click away.
How good is it?	The four protocol/feature-set combinations side by side, including the sub-95% case; the measured train–test gap that answers the overfitting question; per-model accuracy against the 95% threshold; and the confusion matrix with the weakest districts.
What it learned	Global SHAP ranking, per-district top drivers with their direction of effect, and the three-method importance disagreement of §4.3.

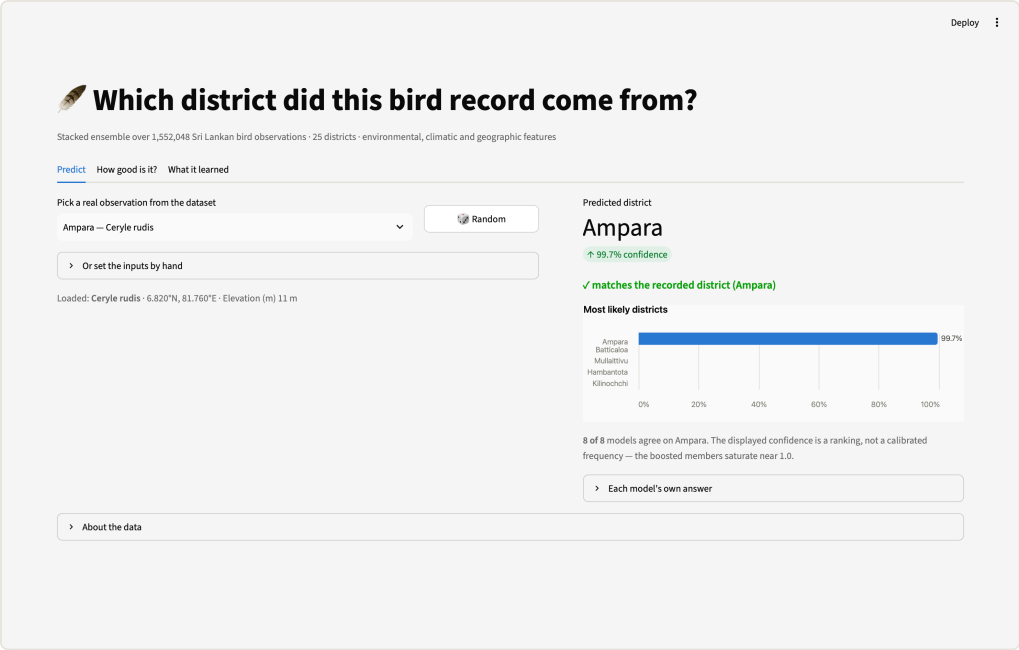


Figure 5. The *Predict* tab: record selection and the collapsed manual inputs on the left; on the right the predicted district, whether it matches the recorded one, and the five most likely districts.

The deployed bundle is a compact build (`n_estimators=40, min_samples_leaf=50`) that fits the free tier's memory budget at **23 MB** on disk, and scores 0.9995 random and 0.9925 grouped — identical to the study-grade build to four decimal places despite its individual forests being measurably weaker, which is itself an argument for ensembling. Forcing single-threaded execution before NumPy is imported and memoising predictions on the feature tuple cut single-row latency from 256 ms to **54.5 ms**.

7 Conclusion

This project applied a stacked heterogeneous ensemble to 1.55 million Sri Lankan bird-observation records. Its most consequential step was taken before any model was fitted: a Bayes-error analysis demonstrated that the original 429-way species framing is capped at 5.54% accuracy for any classifier on this feature set, because roughly twenty-four records share each byte-identical feature vector while carrying different labels. Re-specifying the task as district provenance, whose ceiling on the same features is 100%, produced a well-posed problem the data can genuinely answer.

The ensemble reaches **0.9995** accuracy, **0.9975** balanced accuracy and **0.9980** macro-F1 under the conventional protocol, and that figure was tested rather than asserted: a train–test gap of 0.0005 rules out overfitting, so the score reflects a task that is genuinely easy. The binding limitation is transfer instead — under the strictest guard, unseen sites with coordinates withheld, accuracy falls to **0.9032**, below the 95% threshold, and the gap between those two numbers is the most informative result the project produced. The explainability audit showed the learned structure to be geographically coherent, identifying highland districts by elevation and temperature inversion, coastal cities by SO_2 and night-time radiance, and inland locations by depleted marine aerosol, and recovering the true coordinates of individual districts to within hundredths of a degree. It also produced a methodological finding of its own: three importance methods attribute between 39.4% and 97.9% of total importance to the same two features, and only the ablation resolves the disagreement. With more time the priorities would be spatial block cross-validation to attack the 0.9032 gap, probability calibration to make the displayed confidences trustworthy, and ALE plots in place of partial dependence.

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All metrics in §3 are reproducible from `src/train.py` and all figures in §4 from `src/explain.py`; the exact commands are in the repository README. Random seeds are fixed at 0 throughout. Figures 1–4 were generated on 6,000 held-out test rows; Figure 5 is a screenshot of the deployed application.